

**HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ELECTRICAL AND ELECTRONIC
ENGINEERING**



GRADUATION THESIS

**PEER-TO-PEER (P2P) ENERGY TRADING
FRAMEWORK FOR MULTI-MICROGRID SYSTEMS
USING ADMM-BASED DISTRIBUTED OPTIMIZATION
AND SECOND-ORDER CONE PROGRAMMING (SOCP)**

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Hanoi, June 2025

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Thesis title: *Peer-to-Peer (P2P) Energy Trading Framework for Multi-Microgrid Systems Using ADMM-Based Distributed Optimization and Second-Order Cone Programming (SOCP)*.

- I. Review the state-of-the-art in peer-to-peer (P2P) energy trading within multi-microgrid systems; analyze the role of distributed energy resources (DERs), physical grid constraints, and privacy requirements.
- II. Present the theoretical background on power flow models (from LinDistFlow to SOCP relaxation) and distributed optimization algorithms (ADMM).
- III. Develop a mathematical model for local microgrid energy management, including operational constraints for DGs, BESSs, renewable sources, and the objective function for cost minimization.
- IV. Propose a distributed P2P coordination algorithm combining ADMM consensus with dynamic penalty and SOCP relaxation; derive the locational marginal price (LMP) from ADMM dual variables.
- V. Simulate and analyze results for two case studies (3-MG and 4-MG cross-connected systems); evaluate economic performance, voltage regulation, network losses, and convergence behavior.

Hanoi,// 2025

Supervisor

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ACKNOWLEDGEMENTS

First and foremost, I would like to express my sincere gratitude to **Assoc. Prof. Dr. Nguyen Duc Tuyen** for his dedicated supervision, invaluable guidance, and continuous encouragement throughout the entire course of this thesis. His insightful feedback and deep expertise in power systems have been the cornerstone of this work.

I am deeply grateful to all faculty members of the **Department of Electrical Engineering**, School of Electrical and Electronic Engineering, Hanoi University of Science and Technology (HUST), for imparting knowledge and creating a stimulating academic environment over the past five years.

My heartfelt thanks also go to the members of the **100RE Lab** for their collegiality, knowledge sharing, and moral support during the research process. Working alongside such a motivated group has been both a privilege and an inspiration.

Finally, I owe my deepest appreciation to my **family**, whose unwavering support, trust, and encouragement have been my greatest source of strength and motivation throughout my academic journey.

I acknowledge that the thesis may still contain shortcomings, and I sincerely welcome any constructive feedback from professors and readers.

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ABSTRACT

The rapid penetration of renewable energy sources into distribution networks has driven the emergence of microgrids (MGs) as a fundamental building block of future smart grids. Multi-microgrid (Multi-MG) systems offer significant potential for peer-to-peer (P2P) energy trading, enabling economic dispatch and energy autonomy among prosumers. However, existing coordination frameworks commonly adopt a single-bus MG abstraction, which neglects the physical constraints of the distribution network—including voltage magnitude limits, line thermal limits, and the nonlinear coupling between active power, reactive power, and voltage (P-Q-V). Furthermore, centralized approaches compromise the privacy of individual microgrid operators.

This thesis proposes a fully distributed P2P energy trading framework for multi-microgrid systems that simultaneously addresses both the physical and informational challenges. The framework integrates: (1) a **Second-Order Cone Programming (SOCP) relaxation** of the Branch Flow Model (BFM) to accurately capture multi-bus AC network physics within each microgrid, and (2) an **ADMM (Alternating Direction Method of Multipliers) consensus algorithm with a dynamic penalty parameter** to decompose the global optimization problem into privacy-preserving local subproblems. A key feature of the proposed approach is that the **Locational Marginal Price (LMP)** emerges endogenously from the ADMM dual variables, providing a physically meaningful and cost-reflective pricing signal without requiring a centralized market operator.

Simulation results on two test cases—a three-microgrid radial system and a four-microgrid cross-connected network—demonstrate that the proposed framework achieves superior economic performance, maintains voltage profiles within permissible bounds, minimizes network losses, and converges reliably. Compared to single-bus and linearized (LinDistFlow) benchmarks, the SOCP-based approach eliminates physically infeasible dispatch schedules and provides accurate reactive power coordination.

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DECLARATION OF ORIGINALITY

I, **Trinh Xuan Thanh**, Student ID: [MSSV], Class: Electrical [XX] – K[XX], hereby declare that the work presented in this thesis, entitled “*Peer-to-Peer (P2P) Energy Trading Framework for Multi-Microgrid Systems Using ADMM-Based Distributed Optimization and Second-Order Cone Programming (SOCP)*”, is the product of my own research carried out under the supervision of Assoc. Prof. Dr. Nguyen Duc Tuyen. All data reported are truthful. All cited material complies with intellectual property regulations and is fully referenced. I accept full responsibility for the content of this thesis.

The Author

Trinh Xuan Thanh

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LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Description	Notes
P2P	Peer-to-Peer Energy Trading	—
MG	Microgrid	—
Multi-MG	Multi-Microgrid System	—
DER	Distributed Energy Resource	—
DG	Distributed Generator	—
PV	Photovoltaic System	—
WT	Wind Turbine	—
BESS	Battery Energy Storage System	—
DSO	Distribution System Operator	—
ISO	Independent System Operator	—
ADMM	Alternating Direction Method of Multipliers	Distributed algorithm
SOCP	Second-Order Cone Programming	Convex relaxation
AC-OPF	AC Optimal Power Flow	—
BFM	Branch Flow Model	—
LinDistFlow	Linearized DistFlow	Approximate model
LMP	Locational Marginal Price	Pricing signal
EMS	Energy Management System	—
SOC	State of Charge	(p.u.)

Symbol	Description	Unit / Domain
$\mathcal{N}$	Set of network buses/nodes	–
$\mathcal{L}$	Set of network branches	–
$\mathcal{M}$	Set of microgrids	–
$\mathcal{T}$	Set of scheduling time steps	–
i, j	Bus/node indices	$i, j \in \mathcal{N}$
m	Microgrid index	$m \in \mathcal{M}$
t	Time index	$t \in \mathcal{T}$ (h)

Symbol	Description	Unit
$P_{ij,t}$	Active power flow on branch (i, j) at time t	kW
$Q_{ij,t}$	Reactive power flow on branch (i, j) at time t	kVAR
$\ell_{ij,t}$	Squared current magnitude on branch (i, j)	A ²
$v_{i,t}$	Squared voltage magnitude at bus i	V ²
p_t^m	P2P traded power of MG m at time t	kW
$P_{m,t}^{\text{DG}}$	DG output power of MG m at time t	kW
$P_{m,t}^{\text{PV}}$	PV panel output power of MG m	kW
$P_{m,t}^{\text{WT}}$	Wind turbine output of MG m	kW
$P_{m,t}^{\text{ch}}$	BESS charging power of MG m	kW
$P_{m,t}^{\text{dis}}$	BESS discharging power of MG m	kW
$\text{SOC}_{m,t}$	State of Charge of BESS in MG m at time t	p.u.

Symbol	Description	Unit
r_{ij}	Resistance of branch (i, j)	Ω
x_{ij}	Reactance of branch (i, j)	Ω
ρ^k	ADMM penalty parameter at iteration k	–
λ_t^m	Dual variable / LMP of MG m at time t	\$/kWh
$\bar{V}, \underline{V}$	Upper/lower squared voltage bounds	V ²
$\bar{S}_{ij}$	Apparent power rating of branch (i, j)	kVA
$\eta^{\text{ch}}, \eta^{\text{dis}}$	BESS charging/discharging efficiency	p.u.
$\bar{E}_m, \underline{E}_m$	BESS capacity upper/lower bounds	kWh
ε	ADMM convergence tolerance	–

Danh sách hình vẽ

Danh sách bảng

CHAPTER 1. INTRODUCTION

1.1 Problem Statement: Power System Transition and Energy Coordination Challenges

In the context of global climate change and the depletion of fossil fuels, the traditional power system is undergoing a profound and comprehensive transition. Historically, power systems were designed around large, centralized fossil-fuel power plants that generated electricity and transmitted it over long distances to passive end-users. However, environmental concerns and technological advancements have catalyzed a paradigm shift. The massive penetration of Distributed Energy Resources (DERs)—such as solar photovoltaics (PV), wind turbines, and energy storage systems—has fundamentally altered the operational paradigm of conventional grids [1, 2]. Electricity consumers are no longer passive participants; they have transformed into prosumers capable of self-generating and injecting power back into the grid. This bidirectional flow of energy challenges the traditional unidirectional design of distribution networks, which were not originally built to handle dynamic, localized generation. Consequently, this decentralized evolution poses immense challenges in energy coordination. Integrating millions of DERs requires a flexible, decentralized management mechanism rather than traditional centralized dispatch, to prevent network congestion, manage bidirectional power flows, and avoid safety limit violations within the distribution grid [3].

1.2 From Centralized Grids to Microgrids: The Role of Distributed Energy Resources

To effectively manage the complexity introduced by DERs, the concept of the Microgrid (MG) has emerged as an essential structural solution [4]. Unlike the macrogrid, which operates on a massive scale and is vulnerable to cascading failures, microgrids allow for the localized clustering of loads and distributed generators within a defined electrical boundary. They are capable of operating either in grid-connected mode, drawing or supplying power to the main grid, or in islanded mode, functioning autonomously during grid outages [5]. These entities act as the fundamental building blocks of the Smart Grid, decentralizing energy management, significantly reducing transmission losses over long distances, and enhancing power supply reliability for local communities. Furthermore, the integration of Battery Energy Storage Systems (BESS) into microgrids effectively addresses the inherent intermittency and volatility of renewable energy sources by storing surplus generation during periods of high output and discharging it during peak load hours or times of low generation [6]. As the number of microgrids proliferates across the distribution network, the operational challenge expands from managing a single-microgrid scope to ensuring the optimal coordination and stable operation of intricate, interconnected multi-microgrid systems.

1.3 Peer-to-Peer (P2P) Energy Trading: Enhancing Economic Efficiency and Energy Autonomy

Instead of relying solely on one-way transactions with utility companies via fixed, often rigid pricing mechanisms like Feed-in-Tariffs (FIT), Peer-to-Peer (P2P)

energy trading introduces a disruptive and flexible economic model [7, 8]. P2P trading empowers prosumers and microgrids to directly trade their surplus energy with one another through localized Energy Markets [9]. This peer-driven approach democratizes the energy sector by bypassing traditional utility intermediaries, thereby reducing transaction overheads and empowering consumers. Recent literature demonstrates that, compared to conventional FIT mechanisms, P2P trading yields superior financial benefits for participants, significantly shortens the payback period for initial DER investments, and maximizes the overall social welfare of the participating community [10, 11]. This mechanism not only creates direct economic incentives for sustainable energy investments but also fosters the energy autonomy and resilience of local communities [12]. Nevertheless, realizing P2P trading in practice requires a highly robust mathematical and operational framework to effectively reconcile the economic objectives of diverse market participants with the stringent physical limitations and security constraints of the underlying grid infrastructure.

1.4 Challenges in Multi-Microgrid Networks

1.4.1 Limitations of Single-Bus Microgrid Abstraction

A critical flaw in the majority of existing literature on P2P energy markets is the widespread abstraction of an entire microgrid as a single electrical node (single-bus abstraction) or the assumption of a copper-plate network, where power can flow unhindered between any two points without losses or limits [13, 14]. While computationally convenient, this approach completely neglects the underlying physical constraints of the practical distribution network, such as actual line resistances, inherently radial topologies, and strict branch power capacity limits. When purely economically driven P2P transaction schedules are physically dispatched over the real network, they frequently lead to severe network congestion, critical voltage limit violations, and potential equipment damage [15, 16]. Even when researchers attempt to address this fundamental flaw by utilizing linearized power flow models (e.g., LinDistFlow or DC-OPF), the inherent approximation errors can reach 5-10% in systems with high renewable penetration, compromising the mathematical exactness required for safe and reliable grid operation [17, 18]. Consequently, there is an urgent need to employ rigorous Second-Order Cone Programming (SOCP) relaxations. These advanced techniques transform the highly non-linear and non-convex AC Optimal Power Flow (AC-OPF) problem into a mathematically tractable convex form, guaranteeing global optimality while rigorously preserving the grid's true physical reality and operational safety margins.

1.4.2 Importance of Joint Active-Reactive Power Co-optimization

Modern distribution networks typically exhibit high resistance-to-reactance (R/X) ratios, distinguishing them from transmission systems. Consequently, effective voltage regulation within these localized grids is heavily dependent not only on reactive power (Q) but also substantially on active power (P) flows. Despite this physical reality, many existing P2P trading studies focus exclusively on active power optimization to maximize direct financial returns, completely disregarding the vital reactive power injection and absorption capabilities of the advanced smart inverters coupled with DERs and BESS [19, 20]. Neglecting this crucial aspect severely diminishes the local voltage regulation capabilities, potentially leading to instability, and forces the system to forfeit valuable potential revenue from providing necessary ancillary services to

the main grid. The joint, holistic co-optimization of both active and reactive power for BESS and distributed generators has been conclusively proven to yield significantly higher overall economic efficiency while robustly safeguarding the distribution grid against localized voltage sags and power quality degradation [21–25].

1.4.3 Privacy Requirements and Distributed Pricing

Beyond the complex physical constraints of power flow, safeguarding data privacy represents arguably the most substantial non-technical barrier to the widespread scalability and adoption of interconnected multi-microgrid networks. Traditional Centralized Energy Management Systems (EMS) inherently require the continuous aggregation of highly sensitive operational information from all participating microgrids—including proprietary cost functions, internal network topologies, real-time load profiles, and specific BESS parameters—to a single, remote central server for optimization. This centralized architecture not only blatantly violates the operational privacy of independent microgrid operators but also creates a critical single point of failure, making the entire energy network highly vulnerable to malicious cyberattacks and data breaches [26, 27]. To effectively overcome this critical vulnerability, the Alternating Direction Method of Multipliers (ADMM) algorithm has been widely adopted due to its exceptional ability to preserve absolute data privacy. ADMM achieves optimal coordination by requiring only the iterative exchange of limited boundary power variables and price signals at the point of common coupling, without ever revealing internal parameters [28, 29]. Furthermore, to accurately and fairly reflect the true operational cost of energy at each specific location—explicitly accounting for both physical network losses and potential line congestion—the Distribution Locational Marginal Pricing (DLMP) mechanism, which can be extracted endogenously directly from the ADMM dual variables, proves vastly superior to traditional, overly simplistic uniform pricing schemes [30–32].

1.5 Research Motivation and Objectives

A critical and systematic review of the current literature reveals a clear and persistent research gap: very few studies have successfully integrated a comprehensive, holistic framework that simultaneously encompasses (1) an exact and rigorous physical grid model via SOCP relaxation to ensure operational safety, (2) a fully distributed, privacy-preserving ADMM-based solver featuring dynamic penalty parameters for accelerated algorithmic convergence, and (3) an endogenous, transparent DLMP pricing mechanism specifically tailored for multi-microgrid systems with deeply integrated BESS [33–37].

The primary motivation of this thesis is to definitively bridge this identified gap by developing a robust, unified model for modern decentralized energy networks. The specific research objectives formulated to achieve this overarching goal are as follows:

- To develop a comprehensive, mathematically rigorous energy management model for individual prosumers and microgrids utilizing the Branch Flow Model and SOCP relaxation techniques.
- To design a fully distributed, secure P2P trading algorithm based on the ADMM framework equipped with a dynamic penalty mechanism, thereby guaranteeing 100% operational data privacy and improving computational efficiency.

- To propose and implement an endogenous DLMP mechanism seamlessly derived from the ADMM convergence process to transparently compensate participants for actual network losses and actively alleviate localized grid congestion.
- To extensively simulate, test, and validate the proposed algorithmic performance, critically evaluating both the tangible techno-economic impacts and the convergence stability on complex, interconnected multi-microgrid test systems under various operational scenarios.

CHAPTER 2. THEORETICAL BACKGROUND AND EVOLUTION OF MATHEMATICAL MODELS

2.1 Evolution of Power Flow Models

2.1.1 *Classical Models: Single-Node and Linearized DistFlow*

The development of energy management models for microgrids has historically relied on simplified power flow representations. The most fundamental approach is the single-node abstraction, which aggregates all generations and loads into a single bus [13, 14]. While computationally trivial and suitable for high-level economic dispatch, this model fails to capture internal grid physics. By completely ignoring the network topology, it masks potential line overloads and internal voltage violations, leading to schedules that may be physically unrealizable when subjected to actual network constraints [15, 16]. To introduce spatial awareness without sacrificing linear solvability, the Linearized Distribution Flow (LinDistFlow) model was widely adopted [17, 18]. This model simplifies the power flow equations by ignoring transverse voltage drops and assuming negligible line losses, thereby providing a purely linear approximation of power flows [19, 20]. This linearity was highly attractive as it allowed the use of robust Mixed-Integer Linear Programming solvers. However, its accuracy severely deteriorates in heavily loaded networks or those with substantial reactive power flows. The lossless assumption becomes invalid when bidirectional power flows from high renewable energy penetration increase system losses, rendering it inadequate for modern active distribution networks where precise voltage regulation is mandatory [21, 22, 38, 39].

2.1.2 *Modern Model: Branch Flow Model and SOCP Relaxation*

To overcome the inaccuracies of classical approximations, the exact AC Optimal Power Flow (AC-OPF) modeled via the nonlinear Branch Flow Model (BFM) is required. This exact formulation precisely accounts for line losses, phase angles, and complex power flows. Nevertheless, the non-convex nature of AC-OPF, characterized by quadratic terms and trigonometric relationships, makes it computationally intractable and prone to trapping solvers in suboptimal local optima. Consequently, the Second-Order Cone Programming (SOCP) relaxation has emerged as the state-of-the-art methodology to bridge the gap between physical accuracy and computational feasibility [40]. By relaxing the rigorous nonlinear equality constraint linking active power, reactive power, voltage, and current into a mathematically convex inequality cone, SOCP convexifies the entire problem [19, 41]. This vital transformation guarantees that any local optimum found is also the global optimum, ensuring both global optimality and computational efficiency while rigorously respecting the underlying physics of radial distribution networks [42–44]. Under standard operating conditions—such as avoiding significant reverse power flows and maintaining sensible voltage limits—the SOCP relaxation is mathematically proven to be exact, meaning the relaxed inequality constraint naturally becomes strictly active at the optimal solution, yielding physically meaningful and exact AC results.

2.2 Significance of Reactive Power and Voltage in Distribution Networks

2.2.1 *Nonlinear P-Q-V Coupling in High R/X Ratio Networks*

Unlike transmission grids where the decoupling of active power-angle ($P-\delta$) and reactive power-voltage ($Q-V$) is a valid assumption due to high line reactance, distribution networks possess fundamentally different physical properties, primarily characterized by high resistance-to-reactance (R/X) ratios. In such resistive environments, active power flows cause significant voltage drops, inducing a strong, nonlinear coupling between active power (P), reactive power (Q), and voltage magnitudes (V) [23, 24]. This means that any fluctuation in active power generation from Distributed Energy Resources (DERs) or dynamic power exchanges in peer-to-peer (P2P) markets will simultaneously and directly impact local voltage profiles [45]. Consequently, the traditional paradigm of separating real power scheduling from voltage control is no longer viable. Therefore, energy management systems cannot treat active power trading merely as an isolated economic variable; it must be continuously and rigorously evaluated against its reactive power implications to proactively manage network congestion and maintain strict voltage stability across the entire distribution network [46].

2.2.2 *Trade-off Between Sequential Coordination and Co-optimization*

Traditional energy management frameworks often employ sequential coordination: active power dispatch is solved first to prioritize purely economic minimization, followed by a secondary reactive power adjustment step designed to correct any resulting voltage deviations. This decoupled approach is fundamentally suboptimal because the initial active power dispatch may exhaust the capacity of inverters or push the network state too close to its limits, leaving insufficient margins for the subsequent reactive power adjustments. As a result, this often forces the curtailment of renewable generation or leads to unnecessarily high operational costs to restore grid stability [25, 47]. In contrast, the joint co-optimization of P and Q explicitly integrates both power components into a single objective function and solves their operational constraints simultaneously. By deeply leveraging the often-underutilized reactive power capabilities of Battery Energy Storage Systems (BESS) and smart PV inverters, this holistic co-optimization approach maximizes the overall economic utility of physical assets. It effectively mitigates active power network losses and pre-emptively prevents voltage violations before they occur, proving fundamentally indispensable for the reliable and robust operation of modern multi-microgrid systems.

2.3 Evolution of Grid Optimization Algorithms

2.3.1 *Centralized Optimization and Privacy Barriers*

Historically, multi-microgrid energy management was orchestrated by a centralized Energy Management System (EMS), which actively gathers all detailed network parameters, real-time load demands, internal generator cost functions, and local physical constraints into a single computational hub [26, 27]. While centralized solvers can theoretically and readily identify the true global optimum due to complete system visibility, they exhibit critical real-world vulnerabilities. The strict prerequisite of sharing highly sensitive operational data and proprietary commercial information fundamentally violates the data sovereignty and privacy expectations of independent microgrid

operators, often deterring participation in cooperative frameworks [28, 48]. Furthermore, a fully centralized architecture inherently suffers from severe scalability bottlenecks; as the number of active DERs, smart meters, and dynamic prosumers scales exponentially, the required communication bandwidth and the central computational burden rapidly become unsustainable. This heavy reliance on a central server also exacerbates system vulnerability, introducing a critical single-point-of-failure that could compromise the entire multi-microgrid network during cyber-attacks or communication outages [29, 49].

2.3.2 Decentralized Optimization: ADMM and the Significance of Dual Variables

To resolve the inherent paradox between achieving global mathematical optimality and preserving local data privacy, decentralized optimization algorithms have become a paramount necessity in modern grid operations. Among various distributed techniques, the Alternating Direction Method of Multipliers (ADMM) stands out due to its superior numerical convergence properties, robustness against communication delays, and overall architectural elegance compared to standard dual ascent methods [50–52]. ADMM systematically decomposes the monolithic global coordination problem into smaller, mathematically independent subproblems that can be solved locally in parallel by each individual microgrid EMS. During this iterative process, the only information exchanged between neighboring entities is the expected power flow at their shared boundary nodes (the point of common coupling) alongside a corresponding dual variable, thereby ensuring the absolute privacy of all internal operational parameters and cost functions [53, 54]. Crucially, the dual variables utilized in ADMM are not merely abstract mathematical artifacts necessary for algorithmic convergence; they possess a profound physical and economic meaning. At the point of convergence, these dual variables intrinsically represent the Distribution Locational Marginal Price (DLMP) at the specific boundary nodes [30, 31]. This endogenous DLMP organically acts as a decentralized, dynamic pricing signal that transparently reflects the true, real-time cost of energy provision. By encapsulating not only the marginal cost of generation but also the marginal costs associated with ohmic network losses and line congestion, DLMPs automatically penalize grid-stressing transactions and naturally incentivize energy trading that supports overall system stability [32, 55–57].

CHAPTER 3. MATHEMATICAL MODEL FOR LOCAL MICROGRID ENERGY MANAGEMENT

3.1 Parameters and Notation

3.2 System Model and Operational Constraints

3.2.1 *Bus Power Balance and Network Safety Constraints (SOCP)*

3.2.2 *Distributed Generator (DG) Operational Constraints*

3.2.3 *Battery Energy Storage System (BESS) Operational Constraints*

3.2.4 *Renewable Energy Source (PV and Wind) Constraints*

3.2.5 *P2P Energy Trading Constraints*

3.3 Objective Function: Operating Cost Minimization for Each Microgrid

CHAPTER 4. DISTRIBUTED P2P COORDINATION ALGORITHM BASED ON ADMM AND SOCP

4.1 Global Optimization Problem for the Multi-Microgrid System

4.2 ADMM Consensus Model with Dynamic Penalty Parameter

4.2.1 Local Primal Update

4.2.2 Global Consensus Update

4.2.3 Dual Variable Update and Dynamic Penalty

4.3 Locational Marginal Pricing (LMP) via ADMM Dual Variables

4.4 Proposed Distributed Algorithm

Algorithm 1 Distributed P2P Trading Algorithm: ADMM–SOCP

Require: Network data, DER parameters, initial penalty ρ^0 , convergence threshold ε

Ensure: Optimal schedules $\{P_t^m\}$, LMP signals $\{\lambda_t^m\}$

1: **Initialize:** $\lambda_t^m \leftarrow 0$, $z_t^m \leftarrow 0$, $\rho \leftarrow \rho^0$, $k \leftarrow 0$

2: **repeat**

3: **Step 1 – Local SOCP (Parallel):** Each MG m solves its local SOCP subproblem $\Rightarrow$ updates p_t^m

4: **Step 2 – Global Consensus (DSO):** Aggregate local solutions $\Rightarrow$ update consensus variable z_t^m

5: **Step 3 – Dual Update:** Each MG updates

$$\lambda_t^m \leftarrow \lambda_t^m + \rho^k (p_t^m - z_t^m)$$

6: **Step 4 – Dynamic Penalty:** Update ρ^{k+1} based on primal/dual residual ratio

7: $k \leftarrow k + 1$

8: **until** $\|p_t^m - z_t^m\|_2 \leq \varepsilon$ for all m, t

9: **Extract LMP:** λ_t^m is the nodal price at MG m 's point of connection

CHAPTER 5. SIMULATION AND RESULTS ANALYSIS

5.1 System Data and Simulation Setup

5.2 Case 1: Main Grid Connected to 3 Microgrids

5.2.1 Economic Performance and Algorithm Convergence

5.2.2 Spatiotemporal Power Dispatch and Renewable Integration

5.2.3 Technical Trade-off Analysis: Network Losses and Storage Optimization

5.2.4 Voltage Profile Analysis and LMP Mechanism

5.3 Case 2: Four Cross-Connected Microgrids

5.3.1 Economic Performance and Energy Autonomy

5.3.2 Renewable Integration and Power Coordination

5.3.3 Network Losses, Storage Efficiency, and Voltage Stability

CHAPTER 6. CONCLUSION

LIST OF SCIENTIFIC PUBLICATIONS

- **Trinh Xuan Thanh**, [Co-author(s)], Nguyen Duc Tuyen, “[*Paper Title*]”, [Journal/Conference Name], 2025 (Under Review / Accepted / Published).

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